UNIT 2: CHEMISTRY OF ORGANIC MOLECULES

Biology 9 Mr. Queenan

Text - Ch 2.3 (p 52-56)

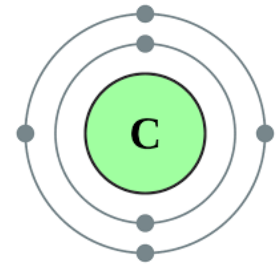
- What makes Carbon special?

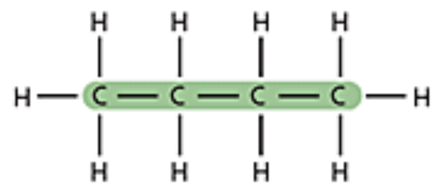
Organic vs. Inorganic Compounds

- All compounds fall into 2 broad categories
  - **Organic**
  - **Inorganic**
- **Organic compounds:** made primarily of carbon atoms
  - Most matter in living organisms consists of organic compounds
- **Inorganic compounds:** generally do not contain carbon

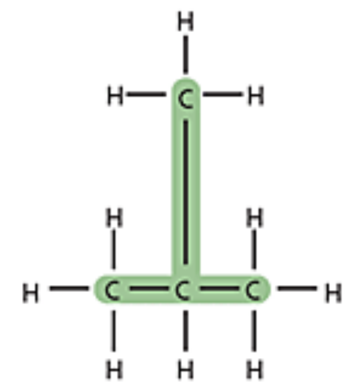
Properties of Carbon

- Carbon:
  - **4 electrons** in valence orbital – needs **8** to be stable
  - Can form up to **4 covalent bonds** with other elements
  - Unlike other elements, can bond with other carbon atoms
    - Forms **straight chains**, **branched chains** or **rings**
    - Results in wide variety of organic compounds

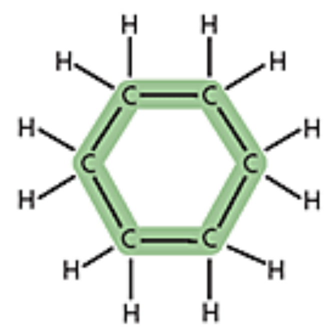




Straight carbon chain



Branched carbon chain

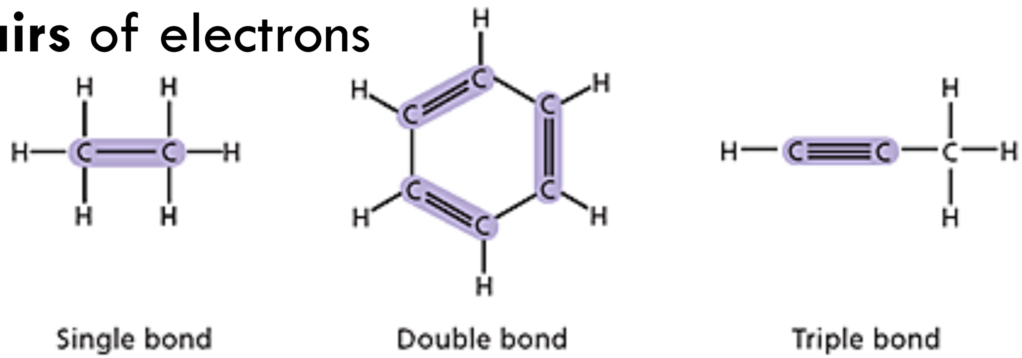


Carbon ring

Simplified view of a carbon ring

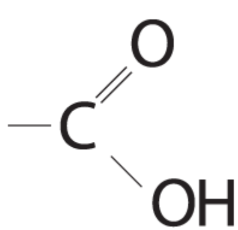
Covalent Bonds

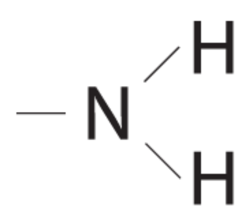
- Types of covalent bonds:
  - **Single bond**
    - Denoted with a **single, solid line** representing **2 elements** sharing **1 pair** of electrons
  - **Double bond**
    - Denoted with a **pair of solid, parallel lines** representing **2 elements** sharing **2 pairs** of electrons
  - **Triple bond**
    - Denoted with **3 solid, parallel lines** representing **2 elements** sharing **3 pairs** of electrons

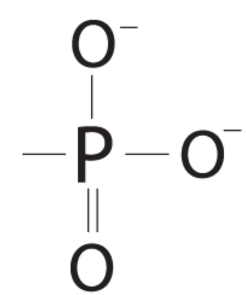


Functional Groups

- **Functional group:** a cluster of atoms attached to an organic compound
  - Influences the characteristics of the molecule
  - Influences the chemical reactions the molecule undergoes
- Ex: **-OH** = **hydroxyl** group
  - Makes the molecule **polar**
  - Makes the molecule **hydrophilic** (water-loving); soluble in water
  - Organic compounds with hydroxyl groups attached known as **alcohols**
- Other functional groups:
  - **Carboxyl**
  - **Amino**
  - **Phosphate**





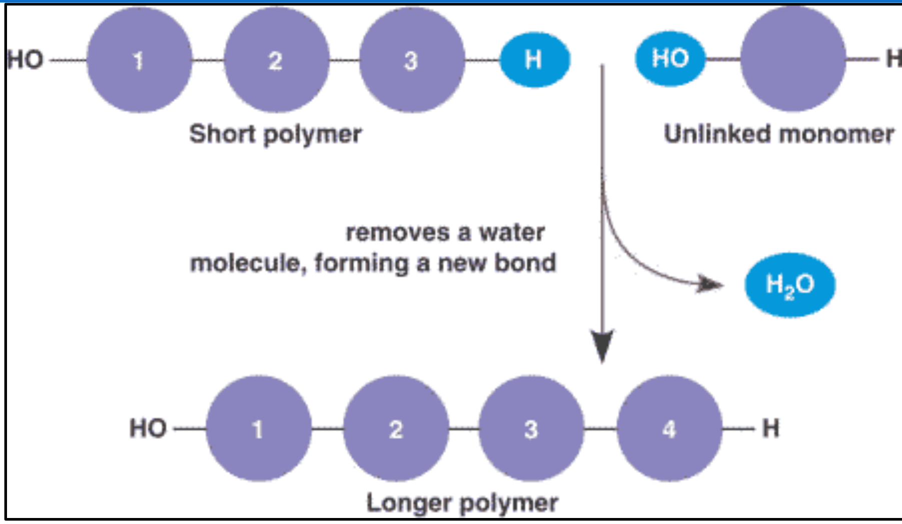


Large C - Molecules

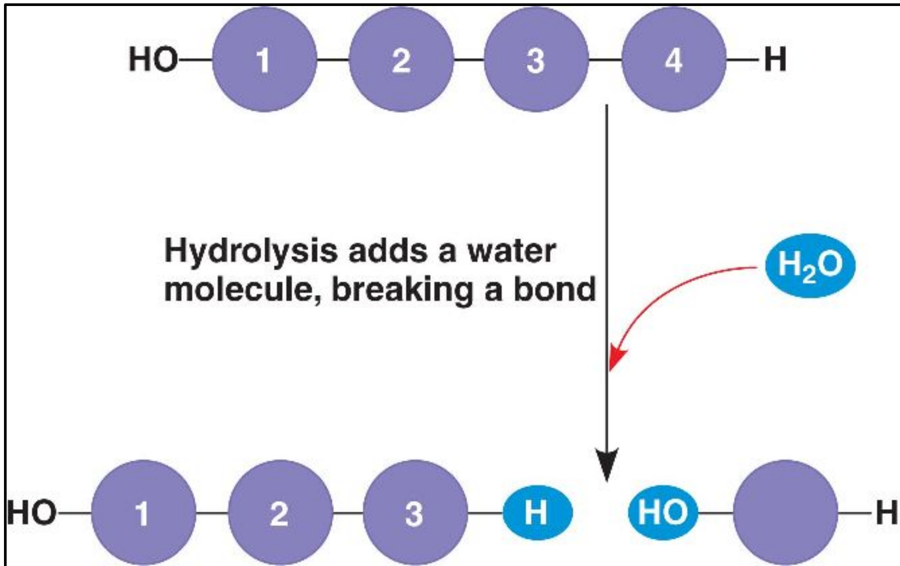
- Many organic compounds built from smaller, simpler molecules called **monomers**
  - Can bond to each other to form **polymers –** a molecule that consists of repeated, linked units
- **Macromolecule –** a large polymer consisting of identical or structurally similar monomers
  - **Carbohydrates**
  - **Proteins**
  - **Lipids**
  - **Nucleic Acids**

Polymer Formation/Degradation

- **Monomers** combine to form **polymers** through a **condensation reaction**
  - Each time a monomer unit is added, a **water** molecule is released
  - AKA **dehydration synthesis**



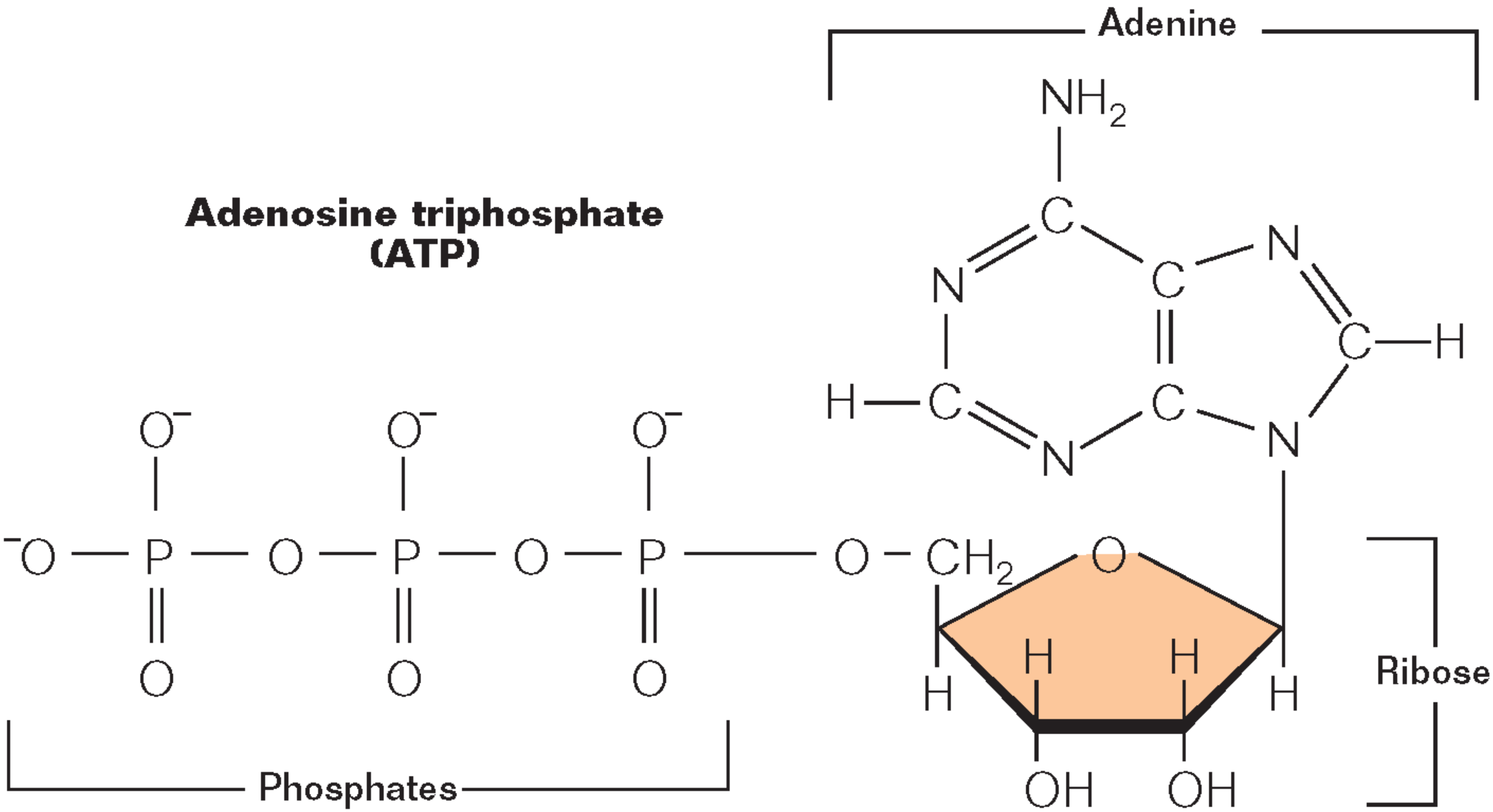
- Polymers are broken down into their component monomers through a **hydrolysis reaction**
  - A **water** molecule breaks the bond linking the monomers



Energy Currency

- Processes of life require constant supply of energy
  - Available to cells in the form of molecules that can **store large amounts of energy in their chemical bonds**
  - Most commonly **adenosine triphosphate, ATP**
    - Composed of 5-carbon sugar (**ribose**), nitrogen-containing double ring (**adenine**), 3 covalently linked **phosphate** groups
- Covalent bonds between phosphates relatively unstable because of their **net negative charge** and **close proximity** to each other
  - Hydrolysis of ATP molecules releases energy that can be used by the cell to drive necessary chemical reactions

ATP



Practice Exercise

- On a sheet of paper using C and H atoms only, draw the following:
  - A 5 carbon structure
  - A 3 carbon structure
  - A 4 carbon structure with a single bonded –OH group

Molecules of Life

1: Carbohydrates

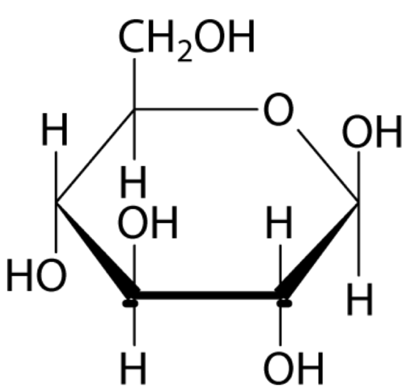
- **Carbohydrates**: Carbon/ Water
  - Organic compounds composed of carbon, hydrogen and oxygen in a ratio of about **1C : 2H : 1O**
  - Number of atoms can vary; ratio remains the same
  - Serve as **energy source** and **structural material**
  - Exist as **monosaccharides, disaccharides** and **polysaccharides**

1: Monosaccharides

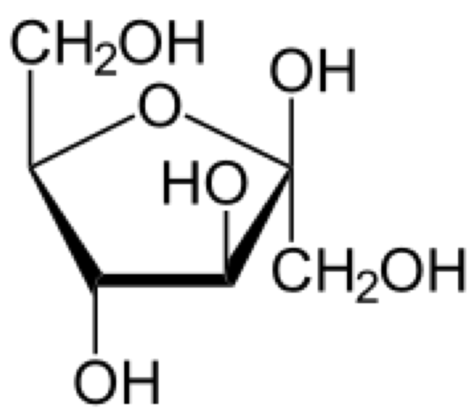
- **Monosaccharide:** monomer unit of carbohydrate
  - Also known as **simple sugar**
  - General formula: $\mathbf{(CH_2O)_{\mathit{n}}}$ where *n* is any whole number from 3-8
    - Ex: $(CH_2O)_6 \rightarrow C_6H_{12}O_6$
- Most common monosaccharides:
  - **Glucose** – energy for cells (what most sugars are broken down into
  - **Fructose** – found in fruit; sweetest simple sugar
  - **Galactose** – found in milk

1: Isomers

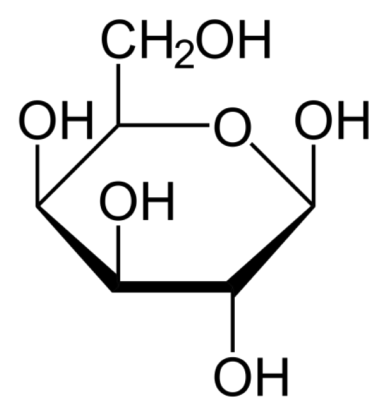
- All have same chemical formula ($C_6H_{12}O_6$) but different structures
  - **Isomer:** compounds that have the same chemical formula, but different structures
- Differences in structure determine the differences in properties of the compounds



Glucose



Fructose

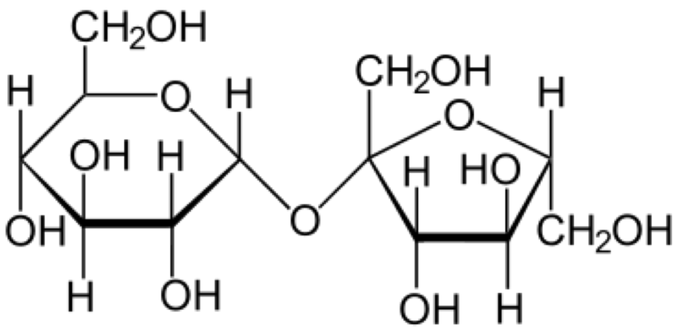


Galactose

1: Complex Sugars

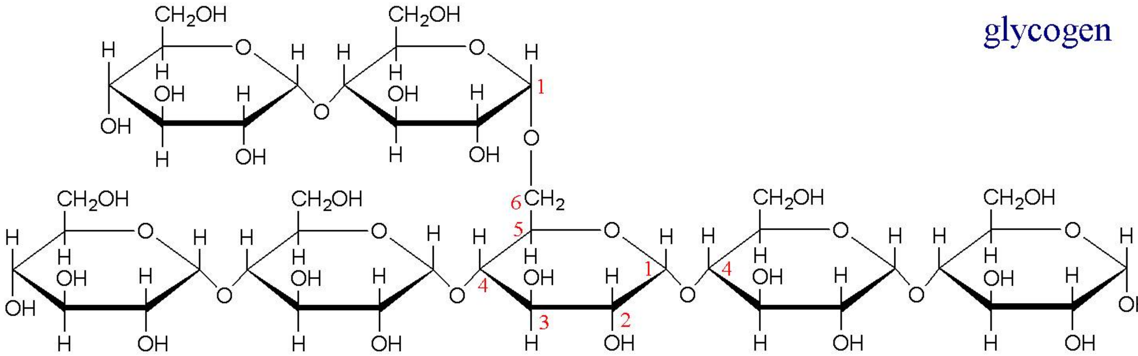
- **Monosaccharides** combine via **condensation reaction** to form **disaccharides** & **polysaccharides**
  - Bond between carbohydrate monomers known as **glycosidic bond**
- **Disaccharide** – double sugar
  - Ex: Glucose + fructose → sucrose (table sugar)
- **Polysaccharide** – complex molecule composed of 3 or more monosaccharides
  - Ex: **glycogen** – animal storage of glucose
    - Hundreds of glucose molecules bonded together in a highly branched chain
    - Stored in liver & muscle; can quickly be broken down into glucose monomers for energy
  - Ex: **starch** – plant storage of glucose
    - Can be branched similar to glycogen or coiled & unbranched
  - Ex: **cellulose** – plant structural molecule; gives strength & rigidity to plant cells; 50% of the composition of wood
    - Thousands of glucose molecules linked in long, straight chains that hydrogen bond to each other resulting in strong structure

1: Di- and Polysaccharide Structures

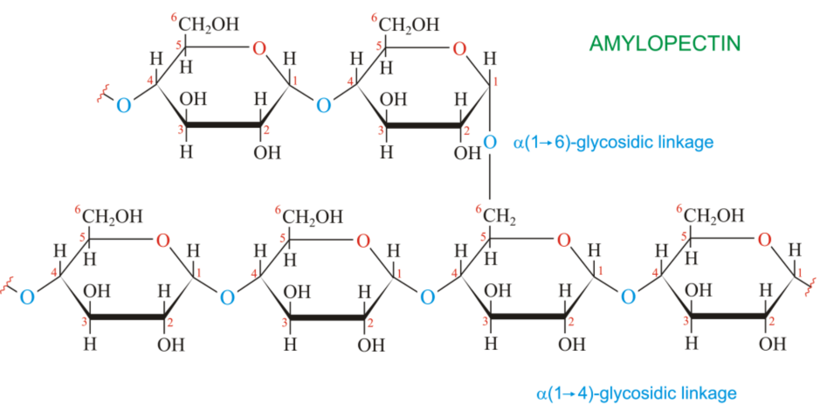


Sucrose

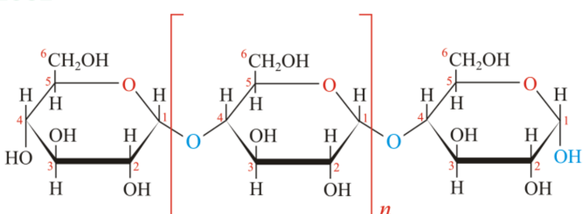
glycogen

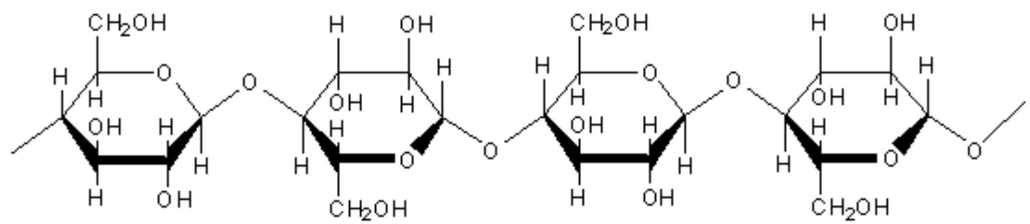


AMYLOPECTIN



AMYLOSE





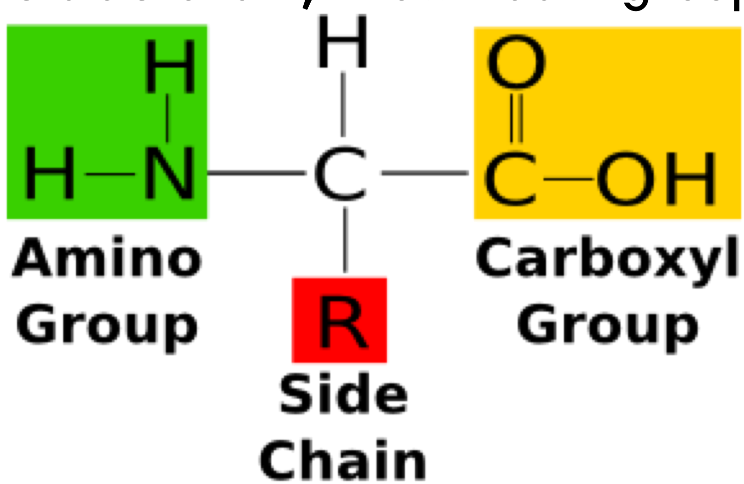
Cellulose

2: Proteins

- **Proteins**: organic compounds composed mainly of **carbon**, **hydrogen**, **oxygen** & **nitrogen**
- Formed by the linkage of monomers: **amino acids**
- Most common molecule in the body (after water)
  - Hair, nails, horns, skin, muscles, enzymes, etc. all composed of proteins
  - 100,000's of protein structures have been identified

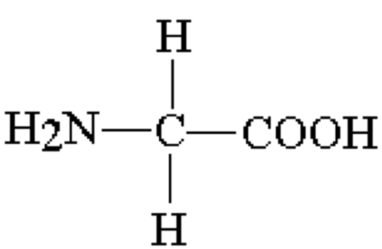
2: Amino Acids

- 20 amino acids; share the same basic structure:
  - Central C atom covalently bonded to 4 other atoms/functional groups
    - Hydrogen atom
    - Carboxyl group
    - Amino group
    - Variable side chain, known as R group

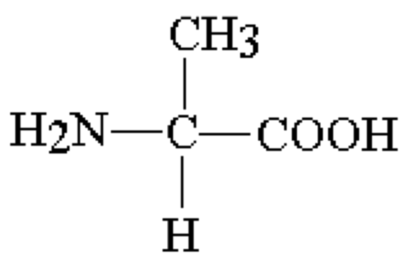


2: Amino Acids

- Amino acids differ in the composition of their R group
  - Can be simple or complex
    - Ex: H (glycine), CH_3 (alanine)
    - Ex: C rings (tryptophan, phenylalanine)

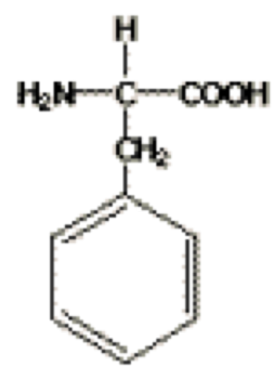


glycine

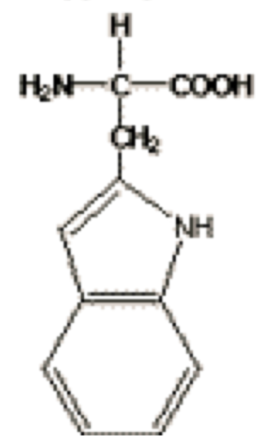


alanine

phenylalanine



tryptophan



2: Importance of R-Groups

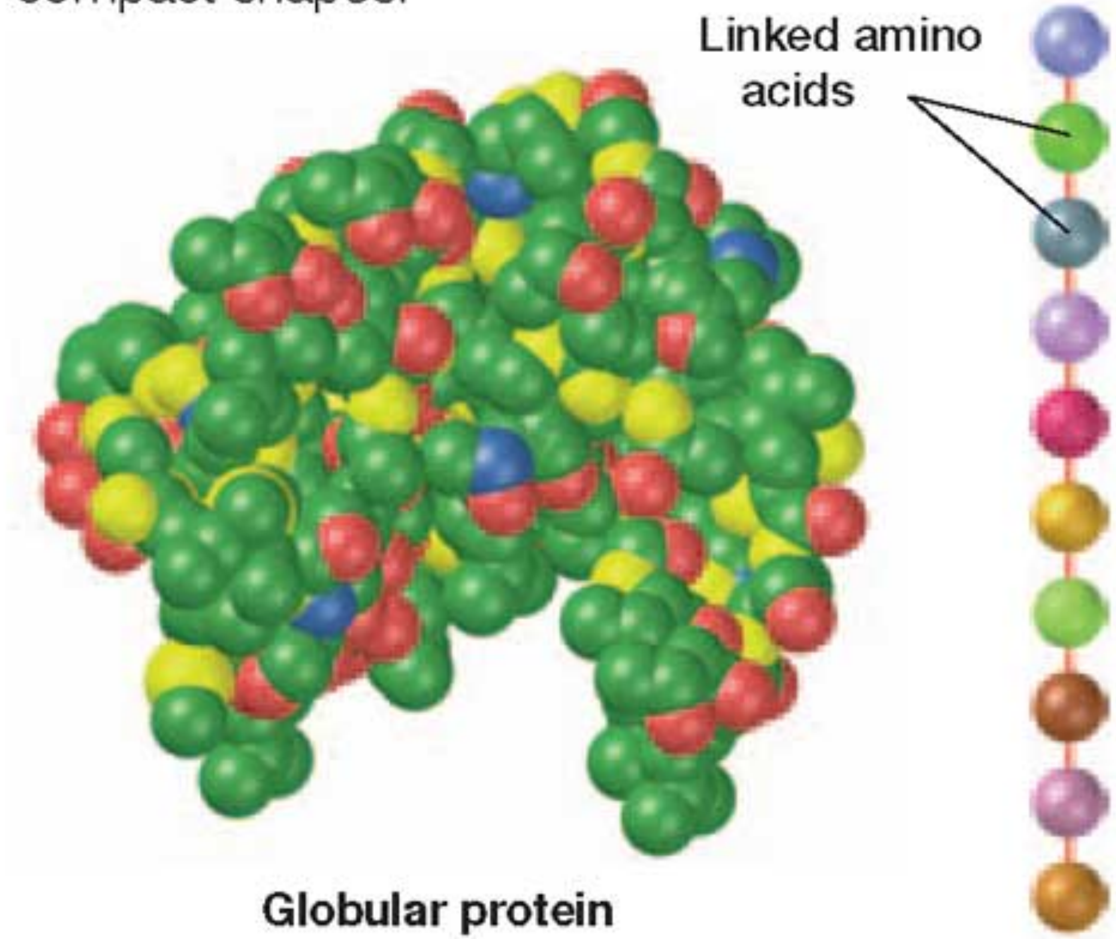
- Differences in R groups gives proteins different shapes/structures
- Different shapes/structures allow proteins to carry out variety of activities in living things
  - Enzymes
  - Bind to and activate/deactivate DNA
  - Inter- and intracellular signaling
  - Cell structures

2: Di- & Polypeptides

- Two amino acids covalently bond to form a **dipeptide**
  - **Condensation** reaction
  - Bond between amino acids known as **peptide bond**
- **Polypeptide:** long chain of amino acids
- Proteins often composed of one or more polypeptides
  - Can be very large, containing hundreds of amino acids
  - Often bend and **fold** on themselves based on interactions between R groups of amino acids
    - Hydrogen bonding
    - Hydrophilic/hydrophobic interactions
- Protein shape also influenced by temperature, pH, solvent, salt content

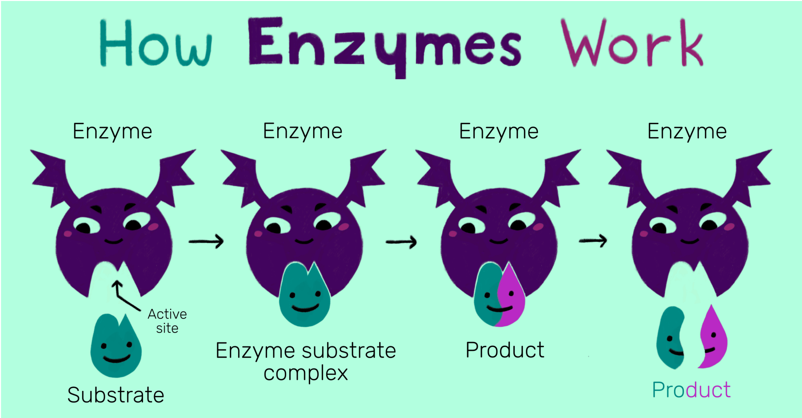
2: Protein Structure

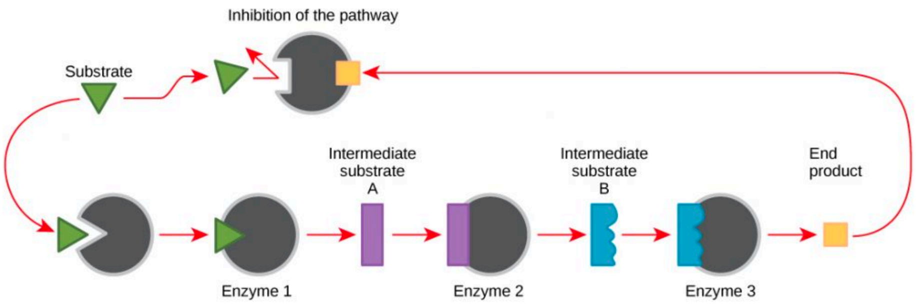
Proteins are chains of amino acids folded into compact shapes.



2: Enzymes

- Enzymes are a special type of protein
  - Biologic Catalyst
    - Work to accelerate chemical reactions
      - In biology, many enzymes act on other proteins known as substrates.
      - Can convert these substrates into other substances, or work to signal chemical pathways within the body to turn on or off.





- We will discuss in greater detail and do a lab assignment with enzymes in a few classes.

3: Lipids

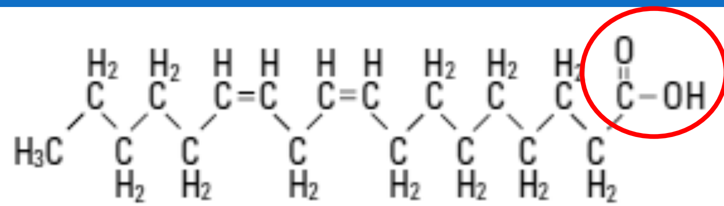
- **Lipids**: large, nonpolar organic molecules
  - Do not dissolve in water
- Includes:
  - **Triglycerides**
  - **Phospholipids**
  - **Steroids**
  - **Waxes**
  - **Pigments**
- Have higher ratio of carbon & hydrogen to oxygen atoms than carbohydrates
  - Higher number of C-H bonds per gram than other organic compounds ; store more energy

3: Fatty Acids

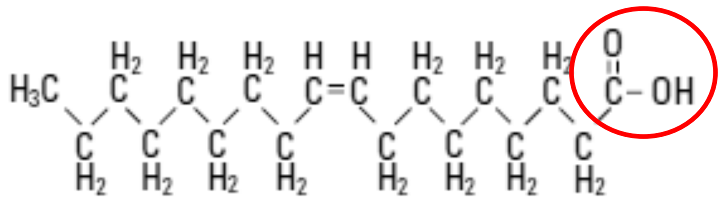
- **Fatty acids:** unbranched carbon chains with a carboxyl group that make up most lipids
  - **Carboxyl end** of molecule is **polar** (hydrophilic) and attracted to water molecules
  - **Hydrocarbon end** of molecule is **nonpolar** (hydrophobic) and repels water molecules
- When carbon is bonded to **4** atoms, it is "**saturated**"
- When carbon forms double bonds within the chain, it is bonded to **less than 4** atoms, and is "**unsaturated**"

3: Fatty Acids

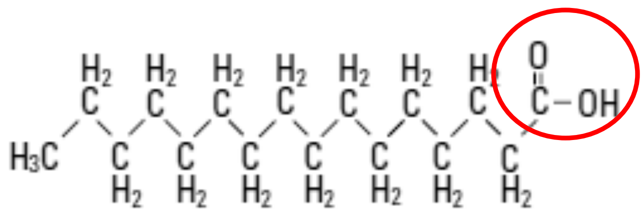
Linoleic acid
$C_{18}H_{32}O_2$



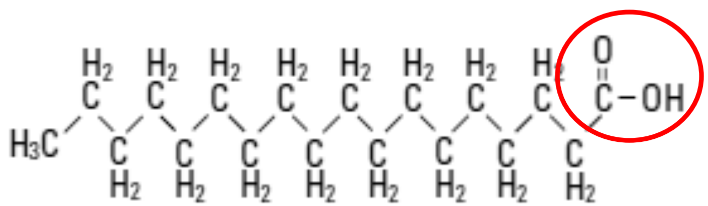
Oleic acid
$C_{18}H_{34}O_2$



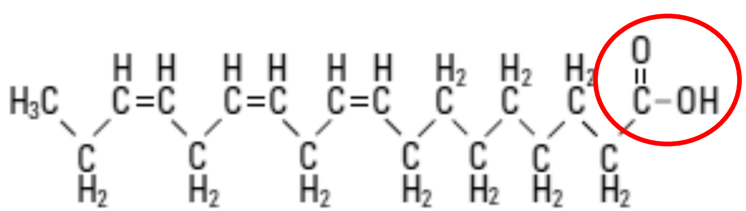
Palmitic acid
$C_{16}H_{32}O_2$



Stearic acid
$C_{18}H_{36}O_2$



Linolenic acid
$C_{18}H_{30}O_2$

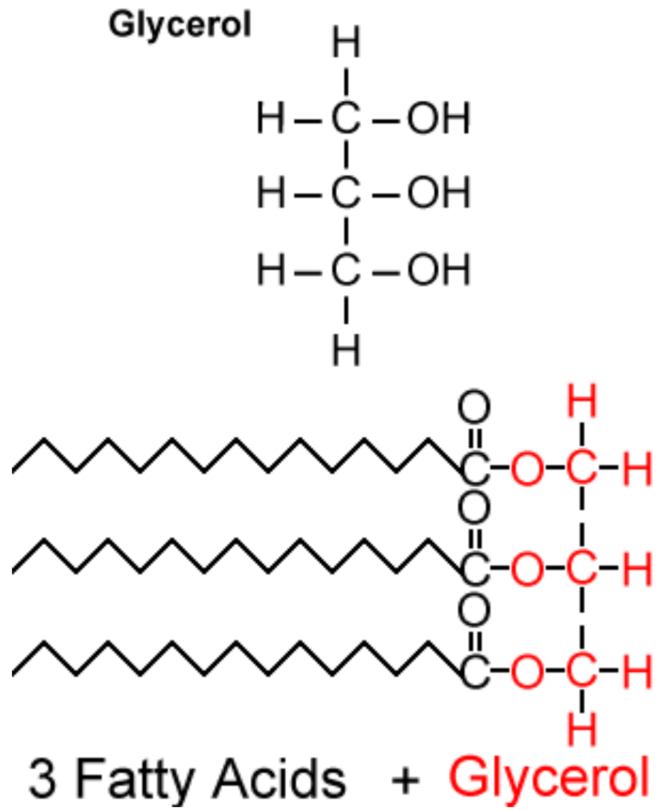


Carboxyl group is the hydrophilic end of the molecule

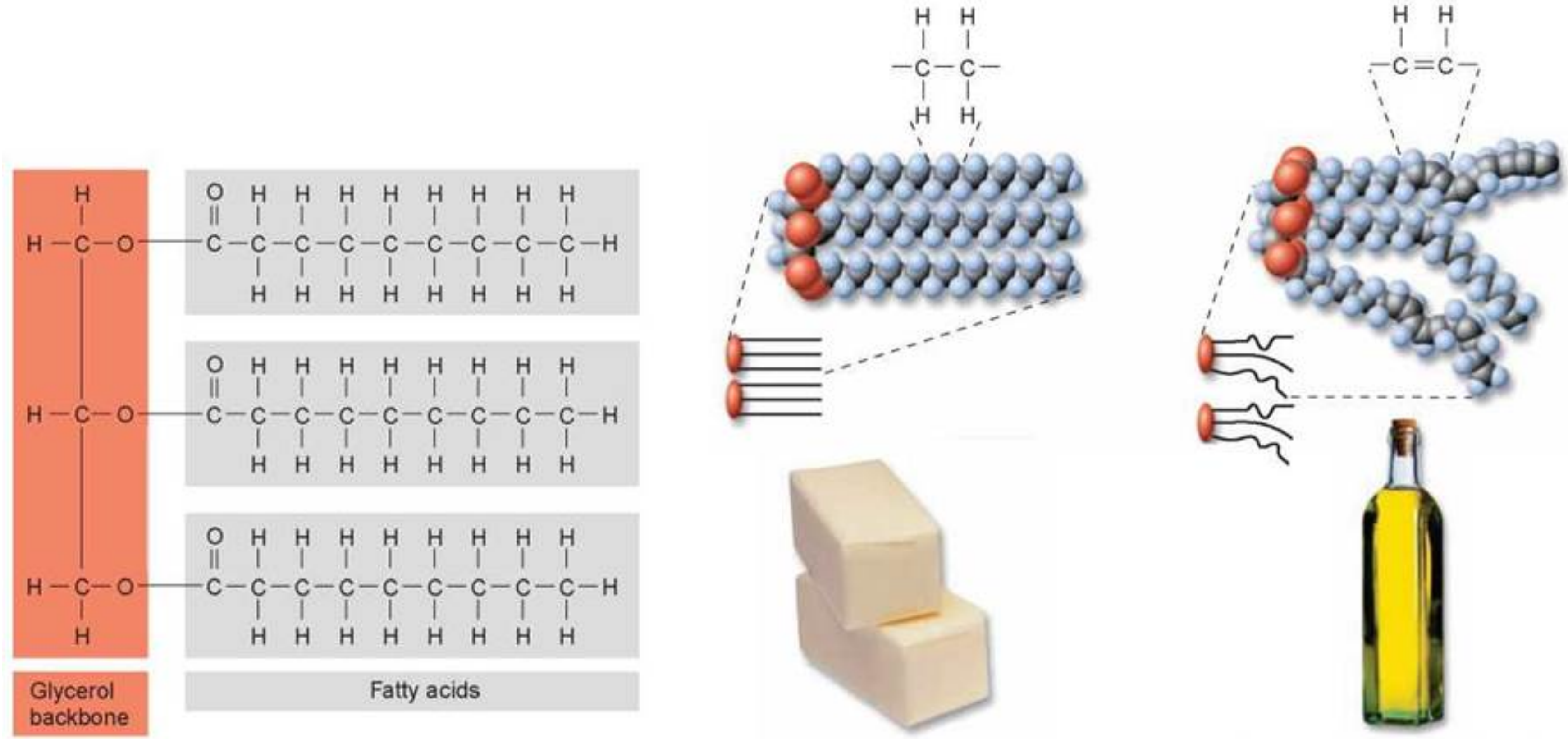
Look at the chemical formula for each of the molecules to the left.
What do you notice about the formulas compared to carbohydrates?

3: Triglycerides

- **Triglyceride:** composed of 3 molecules of **fatty acid** joined to 1 molecule of the alcohol, **glycerol**
  - Saturated triglycerides contain saturated fatty acids
    - Have **high melting points**; tend to be **solid** at room temperature
    - Ex: butter, fat found in meat
  - Unsaturated triglycerides contain unsaturated fatty acids
    - Usually **soft or liquid** at room temperature
    - Found primarily in plant seeds where they serve as energy/C source for germinating plant
    - Ex: olive, corn oil



3: Saturated vs. Unsaturated Triglycerides



(a) Fat molecule (triacylglycerol)

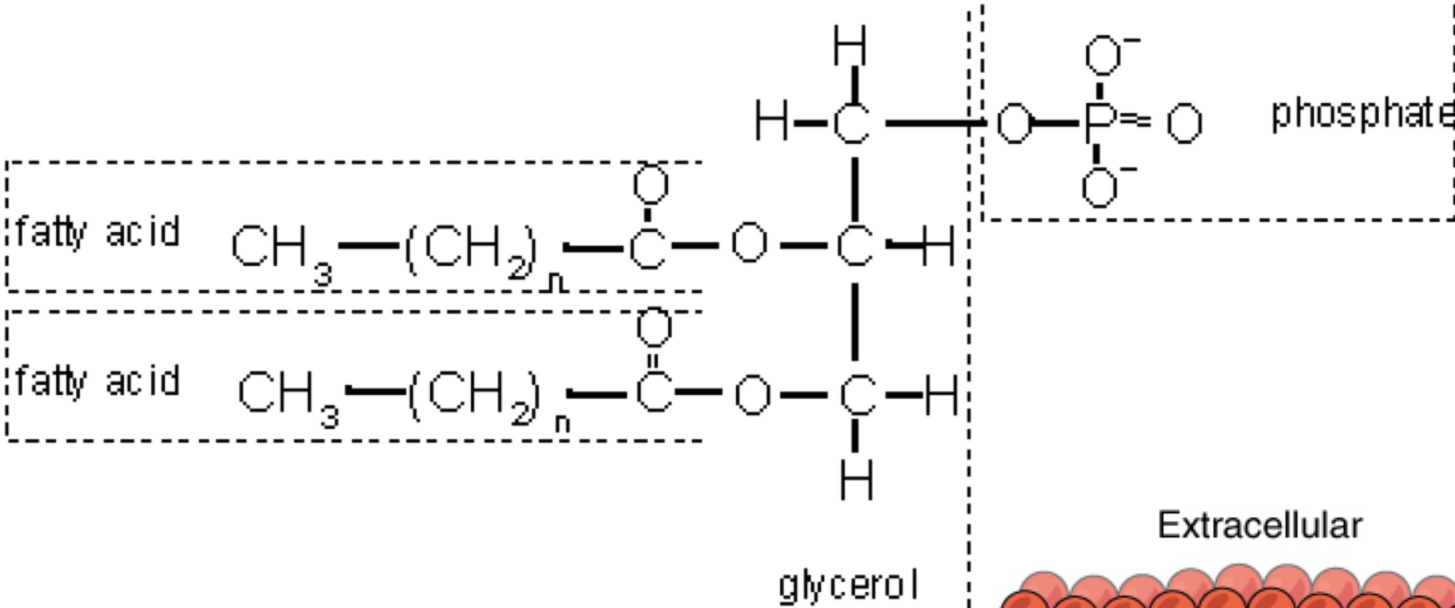
(b) Hard fat (saturated): Fatty acids with single bonds between all carbon pairs

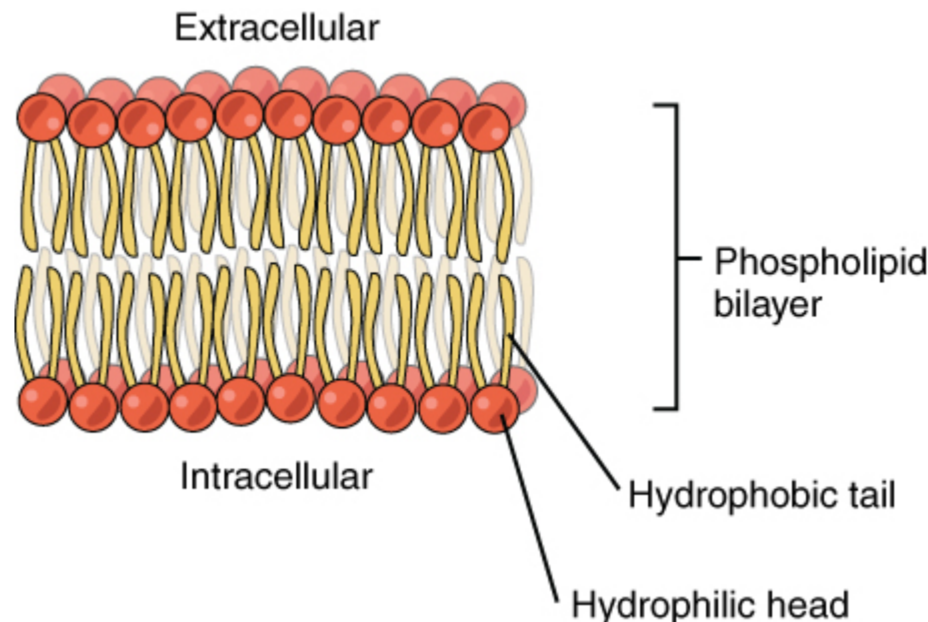
(c) Oil (unsaturated): Fatty acids that contain double bonds between one or more pairs of carbon atoms

3: Phospholipids

- **Phospholipids:** 1 molecule of **glycerol** attached to 2 **fatty acids** and one **phosphate** group
  - **Hydrophilic "head": glycerol/phosphate end** of molecule
  - **Hydrophobic "tail": carbon** chain
- Cell membranes composed of **2 layers** of phospholipids
  - Known as **lipid bilayer**
  - Hydrophobic nature of lipids allow them to serve as barrier between inside & outside of cell
    - Both of which aqueous solutions

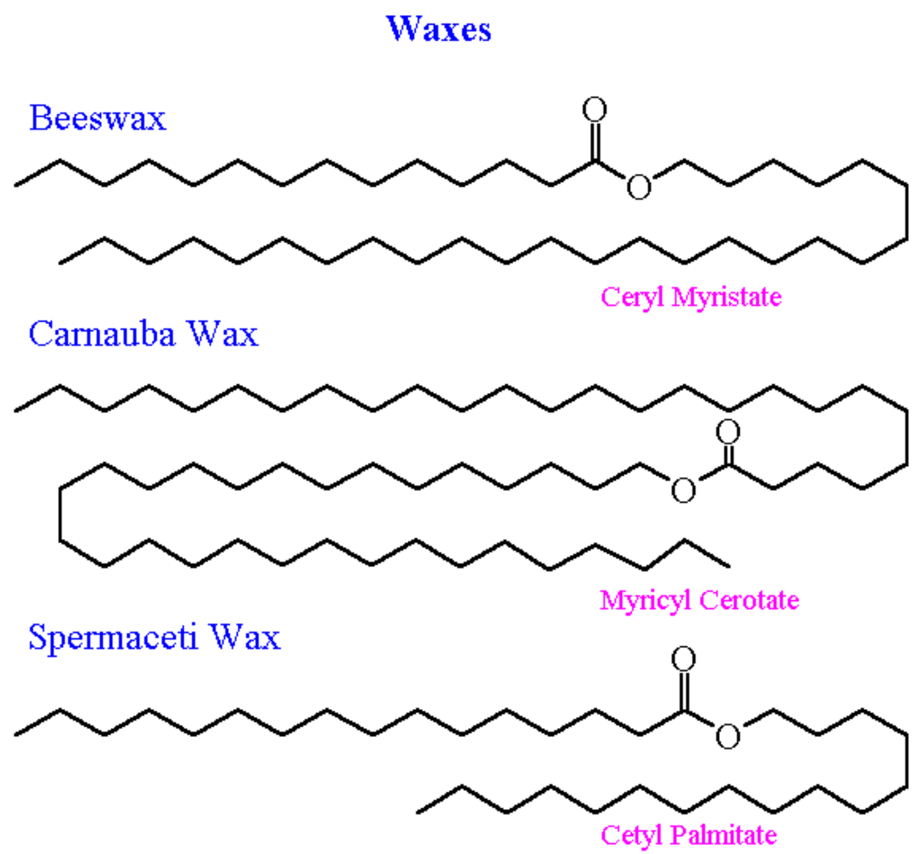
3: Phospholipids





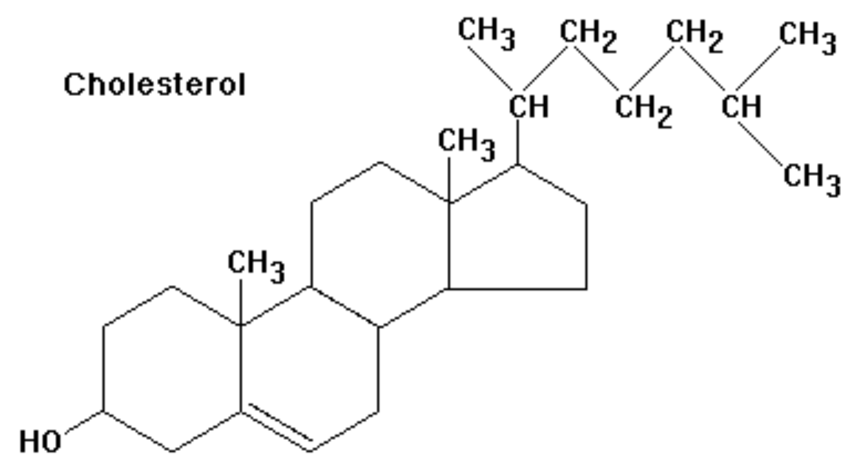
3: Waxes

- **Wax**: type of structural lipid with a long **fatty acid chain** joined to long **alcohol chain**
  - Waterproof
- Plants use waxes as an outer coating for protection of leaves
- Earwax prevents microorganisms from entering the ear canal



3: Steroids

- **Steroid**: molecules composed of **4 fused carbon rings** with various **functional groups** attached
- Many animal **hormones** are steroids
  - Testosterone, growth hormones
- **Cholesterol** is a steroid
  - Necessary for nerve and other cells to function properly
  - Component of the cell membrane
    - Maintains fluidity of the membrane



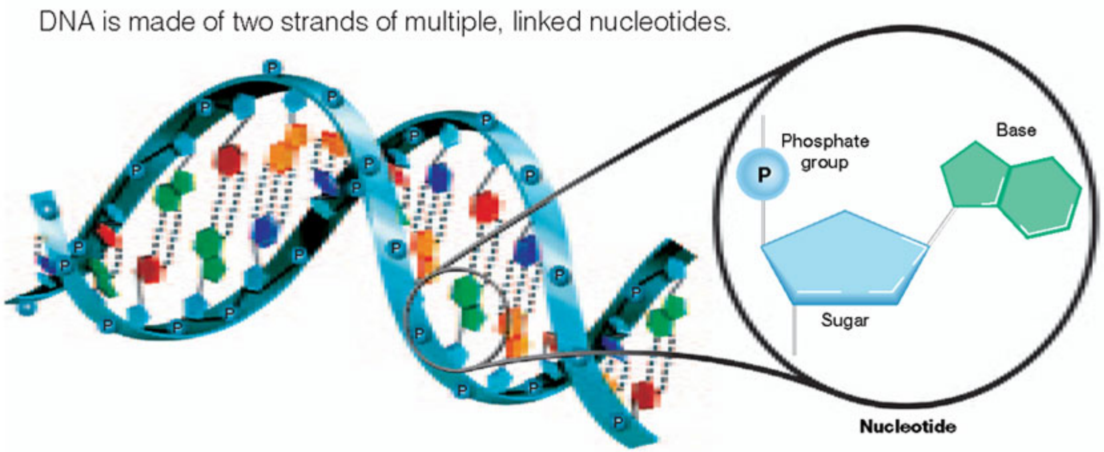
4: Nucleic Acids

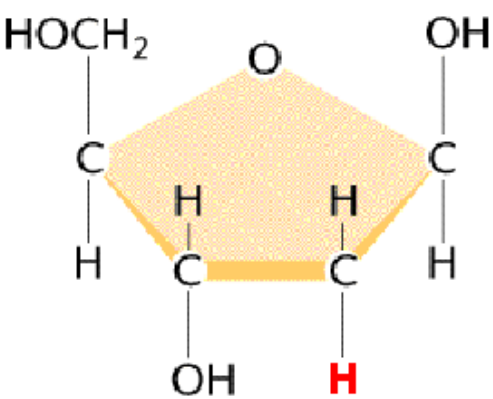
- **Nucleic acid:** very large, complex organic molecules that store and transfer important information in the cell
  - **Deoxyribonucleic acid** (DNA)
  - **Ribonucleic acid** (RNA)
- DNA contains information that determines characteristics of an organism and directs cellular activities
- RNA stores and transfers information from DNA that is essential for manufacturing proteins
  - Can also act as enzymes (ribozymes)
- Polymers composed of thousands of linked monomers known as **nucleotides**
  - Linked by **phosphodiester bond**
  - 3 main components of nucleotide: **phosphate group, 5-carbon sugar, ring-shaped nitrogenous base**

Nucleic Acids

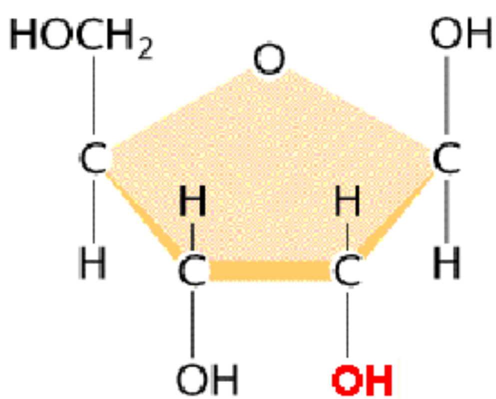
- DNA nitrogenous bases are
  - adenine (A),
  - thymine (T),
  - cytosine (C), and
  - guanine (G).
- RNA also has A, C, and G, but instead of T, it has uracil (U).
- RNA is usually a single polynucleotide strand.
- DNA is a **double helix**, in which two polynucleotide strands wrap around each other.
  - The two strands are associated because particular bases always hydrogen-bond to one another.
  - A pairs with T, and C pairs with G, producing base pairs.

4: Nucleic Acids





2-Deoxyribose



Ribose

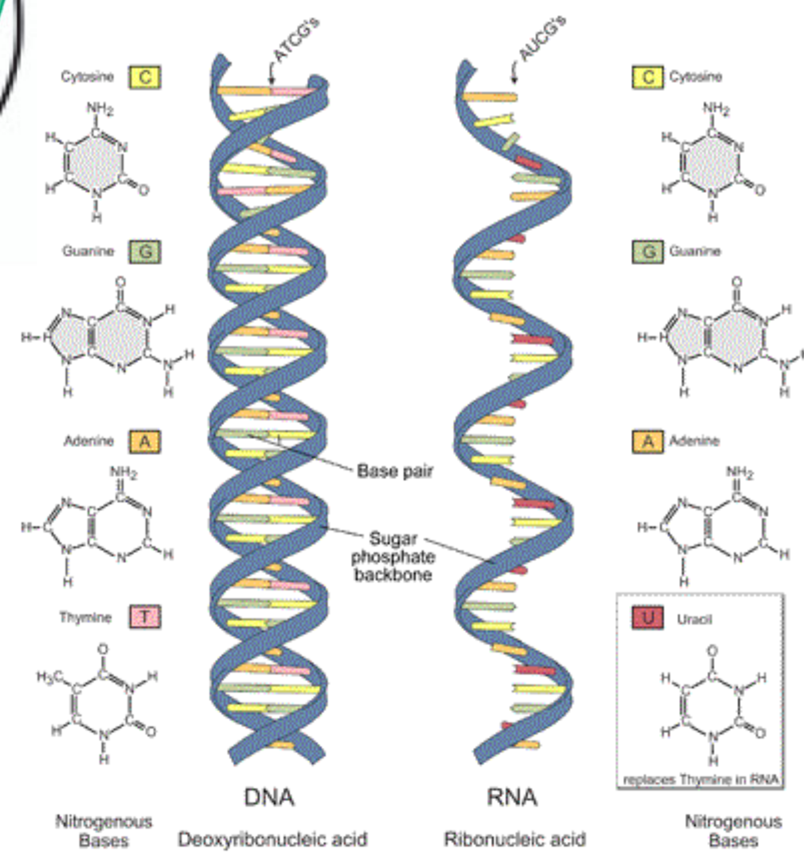
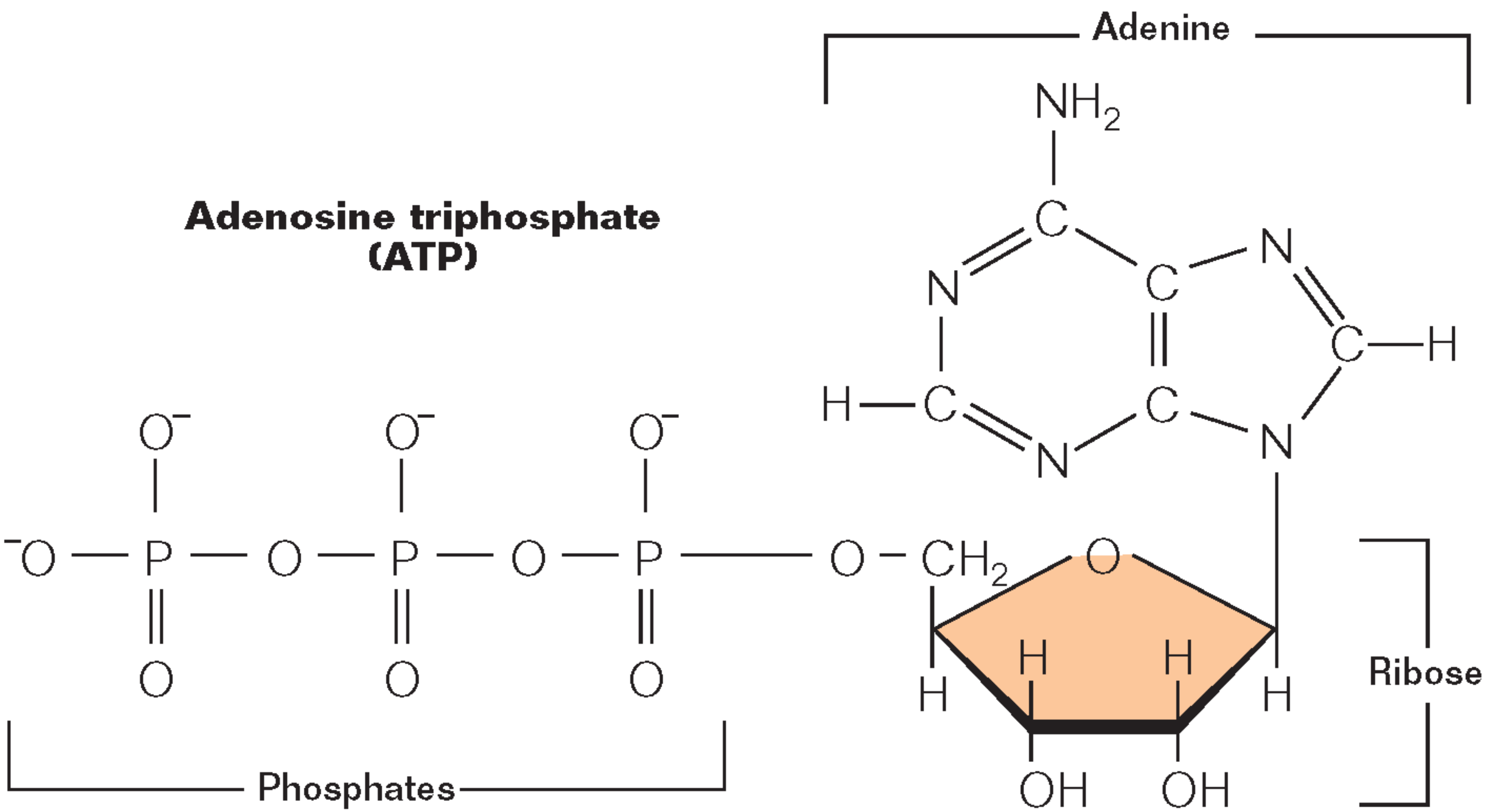


Image adapted from: National Human Genome Research Institute.

4: ATP

Adenosine Triphosphate is a special type of nucleotide, and as discussed already, is a high energy molecule, that powers many cellular processes.



Exit Ticket

- Please go on to Google Classroom and fill out the exit ticket posted for Unit 2.
- For this exit ticket, consider both presentations for Unit 2 to date:
  - Name 3 interesting things you learned so far in Unit 2
  - Name 2 things you want to learn more about in class or on your own
  - Ask one question about a topic that you feel needs more explanation